

Radiotherapy for lung cancer

Radiotherapy may be used to treat small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). It may be used on its own or with other cancer treatments.

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What is radiotherapy?

Radiotherapy is one of the main treatments for lung cancer. You often have it with other treatments. For example, radiotherapy for lung cancer might be given at the same time as chemotherapy. This is called chemoradiation.

Radiotherapy uses high-energy rays to destroy cancer cells. It destroys cancer cells in the area of the body where you have it, while doing as little harm as possible to normal cells.

Radiotherapy is given from a machine that delivers radiotherapy to the cancer. This is sometimes called external beam radiotherapy.

Booklets and resources



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Radiotherapy for lung cancer

When is radiotherapy used to treat lung cancer?

This will depend on the [stage and type of lung cancer](#) and the aim of your treatment.

Your cancer doctor and nurse will talk to you about the aim of your radiotherapy.

Radiotherapy can be given:

- to try to cure the cancer – this is sometimes called radical treatment
- to control the cancer and slow its growth, which can help people to live for longer
- to treat symptoms of advanced lung cancer (palliative radiotherapy).

Radiotherapy for non-small cell lung cancer (NSCLC)

Radiotherapy for [non-small cell lung cancer \(NSCLC\)](#) may be given in the following ways.

- On its own instead of surgery to try to cure [early-stage NSCLC](#). You usually have a type of radiotherapy called stereotactic ablative radiotherapy (SABR).
- After surgery to remove lung cancer to try to reduce the risk of the cancer coming back (adjuvant radiotherapy).
- At the same time as chemotherapy, if the cancer is [locally advanced](#) and you cannot have or do not want lung surgery (concurrent chemoradiation).
- Before or after [chemotherapy](#) if the cancer is locally advanced (sequential chemoradiation).
- To control symptoms if lung cancer has spread to other parts of the body (palliative radiotherapy).

Radiotherapy for small cell lung cancer (SCLC)

Radiotherapy for [small cell lung cancer \(SCLC\)](#) may be given in the following ways.

- Before or after chemotherapy, usually if you cannot have both treatments together (sequential chemoradiation).
- At the same time as chemotherapy (concurrent chemoradiation).
- After chemotherapy if you have [extensive stage SCLC](#) which has responded to treatment (sequential chemoradiation).
- To the head to stop any lung cancer cells that may have spread from growing into a tumour in the brain ([prophylactic cranial irradiation](#), or PCI).
- To control symptoms, if the cancer is more advanced or has spread to other parts of the body (palliative radiotherapy).

Planning your radiotherapy

Before you start treatment, your radiotherapy needs to [be planned](#). This makes sure the radiotherapy is aimed precisely at the cancer and causes the least possible damage to healthy tissue.

Your planning visit

Your first planning visit usually takes 30 to 60 minutes or sometimes longer. You might need more than 1 hospital appointment. Your radiotherapy team will tell you what to expect and let you know if there is anything you need to prepare.

Before the planning visit, some people may need to have a [mould](#) made. This is a light plastic mesh shaped to your chest and shoulders. It fastens to the couch and keeps you still during your treatment. A technician or radiographer usually makes it in the radiotherapy department.

Your planning CT scan

You usually have a planning [CT scan](#) of the chest. This helps your radiotherapy team to work out the precise dose and area of your treatment.

During the scan you lie still on a hard couch. It is important to tell the radiographers if you feel uncomfortable. Once you are in position and comfortable, the radiographers record the details of your position. You will need to be able to lie in the same position for all your radiotherapy sessions.

The radiographers might talk to you about using a breathing technique. This is not suitable for everyone and is not always done at the planning CT scan. It involves holding your breath for short periods of time to help keep your lungs still. You will be asked to use this technique when you have your radiotherapy treatments.

Skin markings

You might have some small permanent markings made on your skin. The marks are about the size of a pinpoint. They help the radiographers make sure you are in the correct position for each session of radiotherapy. Usually, tiny permanent markings are made in the same way as a tattoo.

The marks will only be made with your permission. If you are worried about them, talk to your radiographer.

Having external beam radiotherapy

You usually have external beam radiotherapy as short, daily treatments. These are called treatment sessions or fractions.

Before your first treatment, the radiographers explain what you will see and hear. It is normal to feel a bit nervous. But, as you get to know the staff and understand what to expect, it usually feels less worrying.

Treatment sessions

You might have lung cancer radiotherapy for different lengths of time and in different ways. This can depend on the type of external radiotherapy you have.

Your doctor, [radiographer](#) or nurse will explain how your treatment will be given.

Usually, each radiotherapy appointment takes about 10 to 30 minutes. But you may be in the department for longer. The radiotherapy itself usually only takes a few minutes. Most of the appointment is spent getting you into the correct position and checking your details.

External radiotherapy for non-small cell lung cancer

Some people have standard radiotherapy. This usually means having treatment once a day, Monday to Friday for 4 to 7 weeks, with a break at the weekend.

But there are also some specialised ways of having lung radiotherapy for non-small cell lung cancer (NSCLC).

Stereotactic ablative radiotherapy (SABR) for NSCLC

[SABR](#) is a specialised type of radiotherapy. It may be used to try to cure early lung cancer ([stage 1 to 2](#)).

You might have SABR if you cannot have lung surgery or do not want surgery. SABR is not available at every hospital, so you might have to travel to have it.

The radiotherapy team use scans, specialist machinery and complex planning to target the radiotherapy beams very precisely. This gives a very high dose of radiotherapy to small cancers, and only a low dose to healthy tissue surrounding it. This helps to reduce side effects.

You have fewer treatments over a shorter period. You might have SABR as 3, 5 or 8 treatments usually given every other day. Your doctor or radiographer will explain how many sessions you need over how many days.

SABR might also be used to treat some lung cancers that have spread to other parts of the body.

Continuous hyper-fractionated accelerated radiotherapy (CHART)

If you have [stage 3 NSCLC](#) lung cancer and are not having chemoradiation you might have continuous hyper-fractionated accelerated radiotherapy. This is sometimes called CHART.

You have radiotherapy 3 times a day including weekends, for 12 days. Each treatment must be at least 6 hours apart.

CHART is not available at every hospital, so you may have to travel to have it. You usually have to stay in the hospital or somewhere nearby during treatment.

External radiotherapy for small cell lung cancer

You usually have standard radiotherapy to the lung for small cell lung cancer (SCLC). This might mean having radiotherapy once a day, Monday to Friday, for 4 to 7 weeks.

Radiotherapy can also be given at the same time as chemotherapy. This is called chemoradiation.

Preventive radiotherapy to the brain for SCLC

Your cancer doctor may talk to you about having radiotherapy to the brain. This is called [prophylactic cranial irradiation \(PCI\)](#). With SCLC, there is a risk that cancer cells too small to see on a scan may have spread to the brain. Over time these cells could develop into a secondary cancer in the brain. PCI can reduce this risk and help people with SCLC to live longer.

You usually have PCI after having chemotherapy or chemoradiation which the cancer has responded well to. Your doctor and nurse will explain what is involved and what the side effects are before you decide.

Chemoradiation

Chemoradiation is when chemotherapy and radiotherapy are given at the same time. It is sometimes called concurrent chemoradiation.

Chemoradiation can make treatment more effective. But you usually have more side effects, so you need to be well enough to cope with these.

NSCLC



If the cancer is [locally advanced](#) and surgery is not suitable you may have chemoradiation. You usually start chemotherapy during the first week of radiotherapy. Or you might have 1 cycle of chemotherapy while your radiotherapy is being planned. Some people might have an [immunotherapy drug](#) for up to a year after chemoradiation.

Some people might go on to have [surgery](#) 3 to 5 weeks after chemoradiation. But usually, you will just have chemoradiation.

SCLC



If you have SCLC that has not spread to other parts of the body you might have chemoradiation.

How you have this may depend on the cancer centre where you are having treatment. For example, you may have radiotherapy 2 times a day, Monday to Friday for 3 weeks, along with chemotherapy. If this is too much to cope with, you can have radiotherapy once a day over 4 to 6 weeks.

Some people have chemotherapy before or after radiation. This is called sequential chemoradiation. You might have this because it would be too difficult to cope with both treatments at the same time.

Palliative radiotherapy for lung cancer

Radiotherapy can be given to shrink the cancer and [improve symptoms](#) when lung cancer has spread. This is called palliative radiotherapy. It can be given to different parts of the body, depending on where the cancer has spread.

It may be given to reduce symptoms, such as:

- coughing up blood
- pain in a bone, if the cancer has [spread to the bones](#)
- pain in the chest area
- breathlessness or cough
- increased pressure in the brain, if [the cancer has spread there](#).

Some people have just 1 session of treatment. Other people have it over a few days. Or they might have a higher dose over 1 or 2 weeks. Your cancer doctor or nurse will explain more about this.

[SABR](#) may also be used to treat lung cancer that has spread to other parts of the body such as the liver or the brain.

The side effects of palliative radiotherapy depend on the part of the body being treated. They are usually quite mild.

Side effects of lung radiotherapy

You may get some side effects over the course of your treatment. Your cancer doctor, specialist nurse or radiographer will explain the side effects so you know what to expect.

Always tell them about any side effects you have. There are often things they can do to help. They will also give you advice on how you can manage side effects.

If you have chemoradiation this might make the side effects worse.

Side effects might get worse after radiotherapy finishes. It can take 1 or 2 weeks after treatment before side effects start getting better. After this, most side effects usually slowly go away.

We have more information about the [side effects of radiotherapy](#).

Difficulty swallowing

Radiotherapy can cause inflammation in the gullet (oesophagus). You may have:

- difficulty swallowing or pain when you swallow
- indigestion or heartburn.

These side effects usually happen towards the end of treatment and continue for a few weeks after it finishes. Your doctor can prescribe liquid medicines including pain killers to help reduce the symptoms.

To make swallowing easier, you could try these things:

- Avoid foods that are crunchy or hard to swallow.
- Avoid spicy, hot or very cold foods.
- Avoid citrus fruits like oranges, lemons or grapefruit, or juices.
- Eat softer foods or cut food into small pieces and add sauces or gravy.
- Take painkillers 30 minutes before you eat.
- Eat small meals more often instead of 3 meals a day.
- Drink plenty of fluids – use a straw if it is easier.
- Try not to smoke or drink alcohol – this can make your side effects worse.

If you have difficulty eating, you may meet with a dietitian. They can give you [nutritious, high-calorie drinks](#). You can get these from most chemists, or your GP can prescribe them.

If you are not able to eat or drink enough, your doctor or dietitian might suggest you have [tube feeding](#) to maintain your weight. You have a thin tube that goes through the nose into the throat through which you have fluids and nutrition. This is not often necessary and is only needed until your side effects improve.

Tiredness

Radiotherapy often makes people [feel tired](#). This can build up over your treatment. If you are having other treatments, such as surgery or chemotherapy, you may feel more tired.

It can help to:

- pace yourself and get plenty of rest
- do some light exercise, such as short walks – this will give you more energy.

Sometimes tiredness can continue for weeks or months after treatment finishes. If it does not get better, tell your doctor or nurse.

Skin changes

The skin in the treated area may get dry and itchy. If you have white skin this area might become red. If you have black or brown skin this area might become darker than the surrounding skin.

The radiographers will advise you on how to look after your skin. If it becomes sore, your doctor can prescribe cream to help. It can help to:

- wear loose-fitting clothes made from natural fibres such as cotton
- wash your skin gently with soap and water, gently pat dry and avoid rubbing
- use your usual moisturiser or ask the radiographer for advice
- use your regular deodorant unless the area gets irritated.

When you finish radiotherapy, you should protect the skin in the treated area from direct sunlight. Once any skin reaction has disappeared, use a suncream with a high SPF of 50.

Breathlessness and a cough

You may find your breathing gets worse during radiotherapy and for a few weeks or months after it finishes. You may also get a dry cough. This is because radiotherapy can cause inflammation in the area of the lung being treated. This is called pneumonitis which can be serious.

It is important to contact your cancer team if:

- you have these symptoms
- these symptoms get worse.

They may prescribe steroids to help improve your symptoms. Pneumonitis can happen during radiotherapy or within a few weeks of it finishing.

Blood in your phlegm (sputum)

You might notice small flecks or streaks of blood in your phlegm (sputum). Tell your radiographer, doctor or nurse if this happens. If you are worried about the amount of blood or it increases, let them know straight away.

Hair loss

Hair loss only happens in the treatment area. Men may lose hair on their chest, but it usually grows back. Occasionally [hair loss](#) is permanent.

Late effects of radiotherapy for lung cancer

Late effects are side effects that do not go away, or side effects that develop months or years after treatment.

Some possible late effects after lung radiotherapy include the following:

- Inflammation or scarring (fibrosis) in the treated area of the lung. This can cause breathlessness or a cough. You might find you cannot walk as far after radiotherapy before stopping to catch your breath. If you are worried about your breathing, talk to your cancer team.
- Narrowing of the gullet (oesophagus) making it difficult to swallow. If this happens you may have a small procedure to stretch this narrowing.
- A slight increase in the risk of [heart problems](#), which might cause pain or tightness in the chest.
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Thinning of the bones in the chest area or the back bones, which may cause pain or small breaks (fractures) in these areas.

- Damage to the nerves in the armpit. This is rare, but it can cause pain, numbness or tingling sensations in the arms or hands.
- Lower immunity due to damage to the spleen which is on your left upper side. This might mean you need to have extra vaccinations, and antibiotics to prevent infection.

If you get any of these side effects or others not listed, tell your cancer doctor or nurse straight away. There are different things that can be done to manage late effects. Your doctor will talk to you about the late effects you might get. This depends on the position of the cancer and the type of radiotherapy you have. Also let them know if any side effects do not improve.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our lung cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help